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Linear discrete-time global and regional anti-windup: an LMI approach

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In this article we address linear anti-windup design for linear discrete-time control systems guaranteeing regional and global stability and performance. The techniques that we develop are the discrete-time counterpart of existing techniques for anti-windup augmentation which lead to convex constructions by way of linear matrix inequalities (LMIs) when adopting static and plant order anti-windup augmentation. Interesting system theoretic interpretations of the performance bounds for the non-linear closed loop can also be given. We show here that parallel results apply to the discrete-time case. We derive the corresponding conditions and prove their effectiveness by adapting the continuous-time approaches to the discrete-time case.

Keywords: anti-windup; input saturation; discrete-time

1. Introduction

Anti-windup design characterises augmentations to control systems aimed at recovering stability and performance in light of control input saturation that was not taken into account during the initial controller design. Historically, anti-windup designs arose from practical industrial needs and until recent years the available solutions were mainly intuitive fixes and modifications of the control algorithm in light of activation of saturation (see e.g. Hanus (1988) for a survey of early techniques). Only around the beginning of the 1990s anti-windup augmentation became a theoretical discipline with formal problem statements and precise formulations of stability and performance properties.

Most of the existing literature on anti-windup design applies to the continuous-time case, even though some relevant work has been done in recent years. Among relevant work in the field we find the so-called 'reference governor' or 'command governor' schemes (Bemporad 1998; Angeli and Mosca 1999; Gilbert and Kolmanovsky 1999; Shamma 2000). On the side of classical design approaches, the approaches in Park and Choi (1997); Grimm, Teel, and Zaccarian (2003) provide dynamic linear and non-linear solutions to the discrete-time anti-windup problem. Moreover, in Syaichu-Rohman and Middleton (2004) a linear matrix inequality (LMI) based construction for the static linear anti-windup design is given, while in Gomes da Silva and Tarbouriech (2006) (see also Gomes da Silva and Tarbouriech (2001) for preliminary results in the same direction), a similar construction is derived using a generalised sector condition, so that interesting regional properties can be guaranteed (therefore, the approach becomes also applicable to non-exponentially stable plants). Interesting results are also derived in the twin papers (Turner, Hermann, and Postlethwaite 2003a,b) where a dynamic construction is proposed, which also extends to the sampled-data case. It should be specified that we use the word *regional* in this article to denote any solution guaranteeing non-global results with a guaranteed non-trivial stability region. This fact is significant for anti-windup design, as a matter of fact *local* asymptotic stability always holds by virtue of the fact that the saturation locally acts like an identity. It seems then reasonable to talk about regional properties whenever the stability guarantees are provided in a 'large' region (large in the sense that it is larger than what one would obtain from just doing nothing and relying on the trivial local properties of saturated closed loops).

In this article we address the LMI-based linear anti-windup augmentation problem for discrete-time linear control systems. We fill in here a gap between results available for continuous-time systems and for discrete-time ones. LMI-based linear anti-windup for continuous-time linear systems has been first addressed in Mulder, Kothare, and Morari (2001) where an LMI approach was proposed for the design of static anti-windup augmentation for global closed-loop

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guarantees. As a follow-up, in Grimm et al. (2003a) the approach was extended to dynamic anti-windup augmentation. All these approaches relied on the classical sector condition, which only allowed to derive useful results for exponentially stable plants. Only recently, in Hu, Teel, and Zaccarian (2008) the approach was also extended to the case of regional guarantees by way of generalised sectors. In the discrete-time setting, the counterpart of the work in Mulder et al. (2001) has been studied in Syaichu-Rohman and Middleton (2004), whereas that same approach (namely, static augmentation) has been extended to the case of a generalised sector in Gomes da Silva and Tarbouriech (2006) (thereby guaranteeing useful properties also for non-exponentially stable plants). We develop here a general construction which corresponds to the discrete-time counterpart of our continuous-time papers (Grimm et al. 2003a; Hu et al. 2008). In particular, we give non-linear conditions for the design of a dynamic anti-windup compensator guaranteeing regional or global stability and performance properties and show that these conditions transform into LMIs in the static case and in the plant-order case. For the static case (which had been already characterised in Syaichu-Rohman and Middleton (2004), Gomes da Silva and Tarbouriech (2006)) we are also able to derive feasibility conditions that have a system theoretic interpretation. All the results are parallel to the past results that apply to the continuous-time case. The peculiarity of this discrete-time generalisation, besides its importance in terms of the arising constructive tools, resides in the use of a trick, specific of the discrete-time case (corresponding to Fact 1 in § 5.2), which allows to obtain convex conditions, even though of a higher block dimension as compared to the continuous-time parallel ones. A preliminary version of this article appeared in Massimetti, Zaccarian, Hu, and Teel (2006).

The article is organised as follows: in § 2 we give the problem statement, in § 3 we give tools for analysis of regional and global stability and performance and give feasibility conditions and a constructive procedure for anti-windup compensator synthesis which may optimise certain performance measures. The proofs of the fundamental theorems are reported in § 5.

Notation Given a square matrix X, we denote $\mathrm{He}\ X := X + X^T$. Given a sequence $x(t)$, $t \in \mathbb{Z}_{\geq 0}$, $x^+ = Ax$ is a shorthand notation for $x(t+1) = Ax(t)$. Given matrices $P = P^T > 0$ and H, $\mathcal{E}(P) := \{x : x^T P x \leq 1\}$ and $\mathcal{L}(H) := \{x \in \mathbb{R}^n : |Hx|_\infty \leq 1\}$. A square matrix A is called *Schur stable* if all its eigenvalues are within the open unit disk.

2. Problem definition

2.1 *The unconstrained closed-loop system*

Consider a linear *plant* given by

$$\mathcal{P} \quad \begin{cases} x_p^+ &= A_p x_p + B_{p,u} u + B_{p,w} w \\ y &= C_{p,y} x_p + D_{p,yu} u + D_{p,yw} w \\ z &= C_{p,z} x_p + D_{p,zu} u + D_{p,zw} w, \end{cases} \tag{1}$$

where $x_p \in \mathbb{R}^{n_p}$ is the plant state, $u \in \mathbb{R}^{n_u}$ is the control input, $w \in \mathbb{R}^{n_w}$ is the exogenous input (possibly containing disturbance, reference and measurement noise), $y \in \mathbb{R}^{n_y}$ is the plant output available for measurement, $z \in \mathbb{R}^{n_z}$ is the performance output (possibly corresponding to a weighted tracking error) and A_p, $B_{p,u}$, $B_{p,w}$, $C_{p,y}$, $D_{p,yu}$, $D_{p,yw}$, $C_{p,z}$, $D_{p,zu}$ and $D_{p,zw}$ are matrices of suitable dimensions. The plant with $u \equiv 0$ will be referred to as the *open-loop plant*.

Assume also that an *unconstrained controller* has been designed:

$$\mathcal{C} \quad \begin{cases} x_c^+ = A_c x_c + B_{c,y} y + B_{c,w} w + v_1 \\ y_c = C_c x_c + D_{c,y} y + D_{c,w} w + v_2, \end{cases} \tag{2}$$

(where $x_c \in \mathbb{R}^{n_c}$ is the controller state, $y_c \in \mathbb{R}^{n_u}$ is the controller output, v_1 and v_2 are additional inputs that will be used for anti-windup augmentation and $A_c, B_{c,y}, B_{c,w}, C_c, D_{c,y}$ and $D_{c,w}$ are matrices of suitable dimensions) in such a way that its interconnection to the linear plant through the equations

$$u = y_c, \quad v_1 = 0, \quad v_2 = 0, \tag{3}$$

is well posed and guarantees internal stability of the arising closed-loop system. The interconnection of (1) and (2) via (3) corresponds to the block diagram in Figure 1 which we will refer to as the *unconstrained closed-loop system*. By selecting the state $x_\ell := [x_p^T \quad x_c^T]^T \in \mathbb{R}^{n_{\mathrm{cl}} \times n_{\mathrm{cl}}}$, where $n_{\mathrm{cl}} := n_p + n_c$, and focusing on the effect of the exogenous input w on the performance output z, we can write the dynamics of the unconstrained closed loop system as a single linear system with state space representation

$$\begin{aligned} x_\ell^+ &= A_{\mathrm{cl}} x_\ell + B_{\mathrm{cl},w} w \\ z &= C_{\mathrm{cl},z} x_\ell + D_{\mathrm{cl},zw} w, \end{aligned} \tag{4}$$

where A_{cl}, $B_{\mathrm{cl},w}$, $C_{\mathrm{cl},z}$ and $D_{\mathrm{cl},zw}$ are uniquely determined by the matrices in (1) and (2).

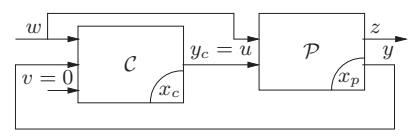

Figure 1. The unconstrained closed-loop system.

2.2 *Input saturation and anti-windup augmentation*

Suppose the control input of the plant is subject to the saturation non-linearity, namely

$$u = \mathrm{sat}(y_c), \tag{5}$$

where sat(·) is the decentralised saturation defined as $\mathrm{sat}(y_c) := [\boldsymbol{\sigma}(y_{c1})\ \boldsymbol{\sigma}(y_{c2}) \ldots \boldsymbol{\sigma}(y_{cn_u})]^T$, where $\sigma(\cdot)$ is the scalar unit saturation function.[1]

Given an integer $n_{\mathrm{aw}} \geq 0$, we address the problem of designing an order n_{aw} linear *anti-windup compensator*

$$\mathcal{AW} \quad \begin{cases} x_{\mathrm{aw}}^{+} & = A_{\mathrm{aw}} x_{\mathrm{aw}} + B_{\mathrm{aw}}(y_c - u) \\ v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} & = C_{\mathrm{aw}} x_{\mathrm{aw}} + D_{\mathrm{aw}}(y_c - u), \end{cases} \tag{6}$$

(where $x_{\mathrm{aw}} \in \mathbb{R}^{n_{\mathrm{aw}}}$ is the anti-windup state, $v \in \mathbb{R}^{n_v}$ (with $n_v := n_c + n_u$) is the anti-windup output, and the matrices A_{aw}, B_{aw}, C_{aw} and D_{aw} are of suitable dimensions) that guarantees a desirable l_2 relationship between the exogenous input w and the performance output z. The interconnection (1), (2), (5) and (6) will henceforth be called the *anti-windup closed-loop system* and is shown in Figure 2.

A desirable stability and performance property for the anti-windup closed-loop system will be presented in terms of Lyapunov analysis tools.

Definition 1: *Given the linear plant $\mathcal{P}$ in* (1) *and the unconstrained controller $\mathcal{C}$ in* (2), *a linear anti-windup compensator* (6) *of order n_{aw} guarantees well-posedness and regional* (*respectively, global*) *quadratic performance of level γ if the augmented anti-windup closed-loop system* (1), (2), (5) *and* (6) *is such that*:

(1) the interconnection (1), (2), (5) and (6) is well posed and
(2) there exists a scalar $\epsilon > 0$ and a quadratic Lyapunov function $V(x) = x^T P x$ (with $x := [\,x_p^T \quad x_c^T \quad x_{\mathrm{aw}}^T\,]^T$ and $P = P^T > 0$) such that its increment ΔV along the dynamics of (1), (2), (5) and (6) satisfies

$$\Delta V \leq -\epsilon x^T x - \frac{1}{\gamma^2} z^T z + w^T w, \tag{7}$$

in a suitable subset of the state space and for all w's with bounded l_2 norm (respectively, for all $x, w \neq 0$).

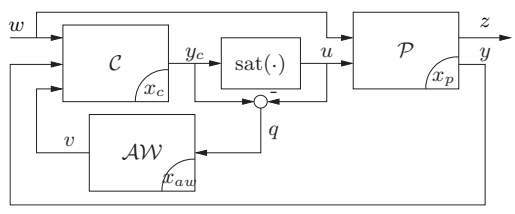

Figure 2. The anti-windup closed-loop system.

Remark 1: Definition 1 entails (sufficient) conditions for regional exponential stability of the anti-windup closed-loop system and for finite l_2 gain γ from w to z for all w's that satisfy a prescribed l_2 bound. Indeed, since the interconnection (5) is well posed (as guaranteed by item 1), item 2 guarantees:

(i) *quadratic stability*, derived by rewriting (7) with $w = 0$, which implies $\Delta V \leq -\epsilon |x|^2$;
(ii) *l_2 gain from w to z smaller than γ*. Indeed, inequality (7) can be summed on both sides from 0 to t (assuming zero initial conditions) to obtain

$$0 \leq V(t) + \epsilon \sum_{\tau=0}^{t} |x|^2 \leq -\frac{1}{\gamma^2} \sum_{\tau=0}^{t} |z|^2 + \sum_{\tau=0}^{t} |w|^2$$

which implies the finite l_2 gain γ from w to z:

$$\| z \| \leq \gamma \, \| w \|_2 \, .$$

3. LMI-based anti-windup analysis and synthesis

The main contribution of this article is presented in three parts. In § 3.1 we will provide tools for regional and global performance analysis when the anti-windup augmentation (6) is preassigned. In § 3.2, we provide non-linear matrix conditions whose feasibility is sufficient to guarantee the existence of an anti-windup compensator that guarantees stability and performance in the sense of Definition 1. For special cases, these non-linear matrix conditions are transformed into a set of LMIs. Finally, in § 3.4, we will give a procedure to construct anti-windup compensators that induce the performance levels guaranteed by suitable solutions to the matrix conditions in § 3.2.

3.1 *LMI-based anti-windup performance analysis*

Assume that the plant $\mathcal{P}$ in (1), the controller $\mathcal{C}$ in (2) and the linear anti-windup compensator $\mathcal{AW}$ in (6) are given. Then, for analysis purposes, the level of performance can be determined by solving an LMI eigenvalue problem.[2]

To formulate suitably the corresponding LMIs, we need to introduce additional notation, which corresponds to representing the anti-windup closed-loop system in a compact way, as in Figure 3. In particular define $\mathrm{d}z : \mathbb{R}^{n_u} \to \mathbb{R}^{n_u}$ with output $q \in \mathbb{R}^{n_u}$ as

$$q = \mathrm{d}z(y_c) := y_c - \mathrm{sat}(y_c). \tag{8}$$

Next, define the overall state variable $x \in \mathbb{R}^n$, where $n := n_p + n_c + n_{\mathrm{aw}}$, as $x := [\,x_p^T \quad x_c^T \quad x_{\mathrm{aw}}^T\,]^T$, which allows the linear dynamics of the plant, controller

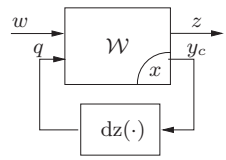

Figure 3. The compact anti-windup closed-loop system.

and anti-windup compensator to be combined and written as

$$\mathcal{W} \quad \begin{cases} x^+ = Ax + B_q q + B_w w \\ y_c = C_y x + D_{yq} q + D_{yw} w \\ z = C_z x + D_{zq} q + D_{zw} w, \end{cases} \tag{9}$$

where the matrices A, B_q, B_w, C_y, D_{yq}, D_{yw}, C_z, D_{zq} and D_{zw} are of appropriate dimensions and are uniquely determined by the matrices in (1), (2) and (6) (see e.g. Hu, Teel, and Zaccarian (2006), Equation (9), for the explicit expressions of the closed-loop matrices).

After a suitable change of coordinates the interconnection between (8) and (9), named the *compact anti-windup closed-loop system* as shown in Figure 3, corresponding to the anti-windup closed-loop system (1), (2), (5) and (6). The following three theorems establish analysis tools for an anti-windup closed-loop system. Their proofs are reported in § 5.1.

Theorem 1: (l_2 *gain for bounded inputs*) *Consider the anti-windup closed-loop* (8), (9) *with* $x(0)=0$. *Given* $\gamma > 0$ *and* $s > 0$, *if there exist a* $Q \in \mathbb{R}^{n\times n}$, $Q = Q^T > 0$, *a diagonal* $U \in \mathbb{R}^{n_u \times n_u}$, $U > 0$ *and* $Y \in \mathbb{R}^{n_u \times n}$ *such that the following holds*:

$$\Phi := \mathrm{He}\begin{bmatrix} -\frac{Q}{2} & 0 & -Y^T & 0 & 0 \\ AQ & -Q/2 & B_q U & B_w & 0 \\ C_y Q & 0 & D_{yq}U - U & D_{yw} & 0 \\ 0 & 0 & 0 & -\frac{I}{2} & 0 \\ C_z Q & 0 & D_{zq}U & D_{zw} & -\frac{\gamma^2 I}{2} \end{bmatrix} < 0 \tag{10}$$

$$\begin{bmatrix} 1/s^2 & Y_i \\ Y_i^T & Q \end{bmatrix} \geq 0, \quad i = 1, \ldots, n_u, \tag{11}$$

(*where* Y_i *denotes the i-th row of* Y) *then* $\|z\|_2 \leq \gamma \|w\|_2$ *for all* w *such that* $\|w\|_2 \leq s$.

Theorem 2: (*Reachable set for bounded inputs*) *Consider the anti-windup closed-loop* (8), (9) *with* $x(0)=0$. *If there exists* $Q \in \mathbb{R}^{n\times n}$, $Q = Q^T > 0$, *a diagonal* $U \in \mathbb{R}^{n_u \times n_u}$, $U > 0$ *and* $Y \in \mathbb{R}^{n_u \times n}$ *such that*

$$\mathrm{He}\begin{bmatrix} -\frac{Q}{2} & 0 & -Y^T & 0 \\ AQ & -Q/2 & B_q U & B_w \\ C_y Q & 0 & D_{yq}U - U & D_{yw} \\ 0 & 0 & 0 & -\frac{I}{2} \end{bmatrix} < 0 \tag{12}$$

and (11) *are satisfied, then*

$$x(t) \in \mathcal{E}(Q^{-1}/s^2), \quad \forall\, t > 0 \tag{13}$$

for all w *such that* $\|w\|_2 \leq s$.

Theorem 3: (*Estimation of the domain of attraction*) *Consider the anti-windup closed-loop* (8), (9) *with* $w \equiv 0$. *If there exist* $Q \in \mathbb{R}^{n\times n}$, $Q = Q^T > 0$, *a diagonal* $U \in \mathbb{R}^{n_u \times n_u}$, $U > 0$ *and* $Y \in \mathbb{R}^{n_u \times n}$ *such that*

$$\mathrm{He}\begin{bmatrix} -\frac{Q}{2} & 0 & -Y^T \\ AQ & -Q/2 & B_q U \\ C_y Q & 0 & D_{yq}U - U \end{bmatrix} < 0 \tag{14}$$

and (11) *are satisfied, then* $V(x) = x^T Q^{-1} x$ *satisfies* $\Delta V(x) < 0$ *for all* $x \in \mathcal{E}(Q^{-1}/s^2) \setminus \{0\}$.

Remark 2: *Non-linear* l_2 *gain*. Given a plant, controller and anti-windup compensator that make up an anti-windup closed-loop system, for each $s > 0$ a relevant problem is that of computing an upper bound on the worst case induced l_2 gain for all w with $\|w\|_2 < s$. This problem can be solved by solving the eigenvalue problem[3] corresponding to minimising γ^2 within the LMI constraints (10) and (11), which provides the tightest bound achievable within the class of quadratic Lyapunov functions. By iterating the computation of the optimal γ^2 for increasing values of $s > 0$ it is possible to construct a curve characterising (an upper bound of) the non-linear gain (see e.g. Megretski (1996)) from the input w to the output z. Similar curves have been computed using the continuous-time counterpart of the conditions derived here in Hu, Teel, and Zaccarian (2006). Paralleling the above mentioned approach, applying Theorems 2 and 3 it is possible to compute via convex optimisation tools suitable for estimates of the domain of attraction and of the reachable set under l_2 bounded inputs for the anti-windup closed loop under consideration.

Although Theorems 1–3 provide useful tools for analysis purposes, they cannot be easily used for anti-windup synthesis because the unknown anti-windup compensator matrices multiply the unknown Q, thus making the matrix inequality (10) non-linear. In the sequel, suitable procedures are given to construct

anti-windup compensators that guarantee well-posedness and quadratic performance.

3.2 *Feasibility of the anti-windup synthesis problem*

To assist in the system theoretic interpretation of the matrix inequalities that will follow, recall the well-known LMI formulation of the bounded real lemma for discrete-time systems.

Lemma 1: (*Bounded real lemma*) *The following statements are equivalent:*

(1) $\|D + C(zI - A)^{-1}B\|_\infty < \gamma$ *and* A *is Schur stable;*
(2) *there exists a symmetric positive definite solution* X *to the LMI:*

$$\mathrm{He}\begin{bmatrix} -X/2 & AX & B & 0 \\ 0 & -X/2 & 0 & 0 \\ 0 & 0 & -I/2 & 0 \\ 0 & CX & D & -\gamma^2 I/2 \end{bmatrix} < 0.$$

The following theorem, representing the main result of this article, provides sufficient conditions for the existence of an anti-windup compensator that guarantees well-posedness, regional quadratic performance and estimates of the domain of attraction and of the reachable set under l_2 bounded inputs. The theorem is proved in § 5.2 and its applications to anti-windup design are investigated in the next sections.

Theorem 4: *Given the plant* $\mathcal{P}$ *in* (1), *the unconstrained controller* $\mathcal{C}$ *in* (2) *and an integer* $n_{\mathrm{aw}} \geq 0$, *if the following set of matrix conditions in the unknowns* (R, S, Z, γ^2) *are feasible:*

$$\mathrm{He}\begin{bmatrix} -\dfrac{R_{11}}{2} & A_p R_{11} + B_{p,u} Z & B_{pw} & 0 \\ 0 & -R_{11}/2 & 0 & 0 \\ 0 & 0 & -I/2 & 0 \\ 0 & C_{p,z} R_{11} + D_{p,zu} Z & D_{p,zw} & -\gamma^2 I/2 \end{bmatrix} < 0 \tag{15a}$$

$$\mathrm{He}\begin{bmatrix} -S/2 & A_{\mathrm{cl}} S & B_{cl,w} & 0 \\ 0 & -S/2 & 0 & 0 \\ 0 & 0 & -I/2 & 0 \\ 0 & C_{cl,z} S & D_{cl,zw} & -\gamma^2 I/2 \end{bmatrix} < 0 \tag{15b}$$

$$R = R^T = \begin{bmatrix} R_{11} & R_{12} \\ R_{12}^T & R_{22} \end{bmatrix} > 0 \tag{15c}$$

$$S = S^T > 0 \tag{15d}$$

$$R - S \geq 0 \tag{15e}$$

$$\mathrm{rank}[R - S] \leq n_{\mathrm{aw}} \tag{15f}$$

$$s^2 R_{11} \geq S_p \tag{15g}$$

$$s^2 R_{11} \leq R_p \tag{15h}$$

$$\begin{bmatrix} 1/s^2 & Z_i \\ Z_i^T & R_{11} \end{bmatrix} \geq 0, \quad i = 1, \ldots, n_u, \tag{15i}$$

then there exists a linear anti-windup compensator of order n_{aw} *that guarantees well-posedness and*

(1) *quadratic performance of level* γ *for all* $\|w\|_2 < s$.
(2) *if* $x(0) = 0$ *and* $\|w\|_2 < s$, *then* $x_p \in \mathcal{R}_p := \{x_p : x_p^T R_p^{-1} x_p \leq 1\}$ *for all* $t \geq 0$.
(3) *if* $w = 0$, *then the origin is exponentially stable and the projection of the region of attraction in the plant direction includes the set* $\mathcal{S}_p := \{x_p : x_p^T S_p^{-1} x_p \leq 1\}$.

Proof: See § 5. □

Remark 3: *On the necessity of the feasibility conditions*. For the global stability and performance case, (Grimm et al. 2003, Theorem 1) is the continuous-time equivalent of our Theorem 4. An important difference between Theorem 4 and Grimm et al. (2003, Theorem 1) is that necessary conditions are also established in the latter while only sufficient ones are proved here. The reason for this difference is that we are also focusing on regional performance and stability properties in this article and, for this, we are relying on improved generalised sector conditions which only apply to decentralised saturation functions. This fact does not allow us to pursue the same approach used in Grimm et al. (2003) where the classical sector condition was used and the whole set of non-linearities in the sector was characterised to establish the necessity part of the proof. When restricting the attention to global properties, the same approach as in Grimm et al. (2003) could be followed also here and the conditions in Theorem 4 would also become necessary.

Remark 4: *Optimal anti-windup design*. The goal of optimal anti-windup design is to construct an anti-windup compensator that optimises one of the three features described in Theorem 4. This can be accomplished by solving a non-convex optimisation problem which optimises one of the following three parameters (possibly imposing a certain level of guaranteed performance on the two other ones) subject to the constraints (15): minimisation of the l_2 gain γ^2,

maximisation of the size of the region of attraction (by maximising S_p) and minimisation of the reachable set under bounded inputs (by minimising R_p).

Remark 5: *Lower bounds on performance level*. Using Lemma 1, conditions (15a) and (15b) have a system theoretic interpretation. In particular, observe that condition (15a) constrains γ to be not less than the $\mathcal{H}_\infty$ norm of the plant $\mathcal{P}$ with a static state feedback given by $K_p = ZR_{11}^{-1}$. Note, however, that this state feedback cannot be selected arbitrarily since constraints (15i) establish a coupling condition between Z and R_{11}. Note that larger selections of Z will retain feasibility of the constraints only for smaller selections of s, namely, the performance properties will be guaranteed in smaller and smaller subsets of the state space (a similar result applies to the continuous time case investigated in Hu et al. (2008)). As s approaches 0 (so that global properties are guaranteed), Z will also approach zero and γ will approach the l_2 gain from w to z associated with the open-loop plant $\mathcal{P}$, mimicking the corresponding condition in the continuous time case, characterised in Grimm et al. (2003, Remark 7). Similarly, paralleling the continuous-time case commented in Grimm et al. (2003, Remark 7), condition (15b) constrains γ to be no less than the l_2 gain of the unconstrained closed-loop system (4). While these two LMIs provide lower bounds for the l_2 gain achievable by the anti-windup closed-loop system, conditions (15e) and (15f) establish a non-linear coupling between the two conditions.

Based on the previous remark, it is evident that when seeking for global results (so that $s = +\infty$ and consequently $Z = 0$) condition (15a) is only feasible if the plant (1) is exponentially stable. It follows that, paralleling the continuous-time results in Grimm et al. (2003), for anti-windup feasibility, asymptotic stability of the plant is necessary if one wants to guarantee the global properties of Definition 1. One of the reasons that are necessary for A_p to be Schur stable is that we are asking for global quadratic stability in the absence of inputs. Even if we did not insist on quadratic stability, with appropriate detectability and stabilisability conditions from w to z, it is a straightforward consequence of the classical small gain theorem that finite gain l_2 stabilisability by bounded controls implies that A_p is Schur stable. In the more general case of non-asymptotically stable linear plants, the global properties must be relaxed (by allowing non-zero selections of Z) to be able to guarantee useful results.

In the next section, we will show that the non-linear condition (15f) can be transformed into a linear one, in some special cases.

3.3 *LMI formulations of the feasibility condition*

A useful property of Theorem 4 is that all but one of the conditions (15) are linear with respect to the unknowns (R, S, Z, γ^2), the exception being condition (15f) – the rank condition. Paralleling the necessary and sufficient conditions for reduced order $\mathcal{H}_\infty$ control synthesis (see e.g. Gahinet and Apkarian (1994), Equation (26)), when considering the *full order* case $n_{\mathrm{aw}} = n_p + n_c$, the rank condition is trivially satisfied and the optimisation of the performance level and the determination of the corresponding solution reduces to a convex LMI eigenvalue problem.

For the full order case, the rank condition is guaranteed satisfied and the optimal performance level $\overline{\gamma}$ such that (15) are feasible can be determined by solving in the unknowns (R, S, Z, γ^2) the LMI eigenvalue problem which may optimise the (squared) performance level γ^2, the reachability set $\mathcal{R}_p$ or the region of attraction $\mathcal{S}_p$. However, when considering anti-windup compensation of *reduced order* ($n_{\mathrm{aw}} < n_p + n_c$), the rank condition needs to be satisfied and the conditions (15) becomes non-linear. By exploiting the special structure of the anti-windup design problem, in the following Propositions 1 and 2 we will show how to replace the non-linear rank condition with equivalent linear conditions, for the special reduced order cases $n_{\mathrm{aw}} = 0$ and $n_{\mathrm{aw}} \geq n_p$, respectively. In these two special cases, all the matrix inequalities are linear in the unknowns, and the optimisation problem becomes a convex LMI eigenvalue problem. To simplify the exposition, we remove in the following statements the constraint (15g) and (15h) characterising a guaranteed stability region and reachability region and focus on the l_2 gain performance measure.

Proposition 1: ($n_{\mathrm{aw}} = 0$) *Given the plant* $\mathcal{P}$ *in* (1), *the controller* $\mathcal{C}$ *in* (2), *conditions* (15) *with* $n_{\mathrm{aw}} = 0$ *are feasible if and only if the following LMI conditions are feasible:*

$$\mathrm{He}\begin{bmatrix} -\frac{R_{11}}{2} & A_pR_{11} + B_{p,u}Z & B_{pw} & 0 \\ 0 & -R_{11}/2 & 0 & 0 \\ 0 & 0 & -I/2 & 0 \\ 0 & C_{p,z}R_{11} + D_{p,zu}Z & D_{p,zw} & -\gamma^2 I/2 \end{bmatrix} < 0 \tag{16a}$$

$$\mathrm{He}\begin{bmatrix} -R/2 & A_{\mathrm{cl}}R & B_{cl,w} & 0 \\ 0 & -R/2 & 0 & 0 \\ 0 & 0 & -I/2 & 0 \\ 0 & C_{cl,z}R & D_{cl,zw} & -\gamma^2 I/2 \end{bmatrix} < 0 \tag{16b}$$

$$R = R^T = \begin{bmatrix} R_{11} & R_{12} \\ R_{12}^T & R_{22} \end{bmatrix} > 0 \tag{16c}$$

$$\begin{bmatrix} 1/s^2 & Z_i \\ Z_i^T & R_{11} \end{bmatrix} \geq 0, \quad i = 1, \ldots, n_u. \tag{16d}$$

Proof: If $n_{\mathrm{aw}} = 0$, condition (15f) is satisfied if and only if $R = S$; thus (15e) is satisfied and (15d) is redundant. Hence, the proof follows by rewriting the remaining inequalities in (15) with $R = S$. □

Proposition 2: ($n_{\mathrm{aw}} \geq n_p$) *Given the plant $\mathcal{P}$ in* (1), *the controller $\mathcal{C}$ in* (2) *and an integer $n_{\mathrm{aw}} \geq n_p$, condition* (15) *are feasible if and only if the following LMI conditions are feasible:*

$$\mathrm{He}\begin{bmatrix} \frac{R_{11}}{2} & A_p R_{11} + B_{p,u} Z & B_{pw} & 0 \\ 0 & -R_{11}/2 & 0 & 0 \\ 0 & 0 & -I/2 & 0 \\ 0 & C_{p,z} R_{11} + D_{p,zu} Z & D_{p,zw} & -\gamma^2 I/2 \end{bmatrix} < 0 \tag{17a}$$

$$\mathrm{He}\begin{bmatrix} -S/2 & A_{\mathrm{cl}} S & B_{cl,w} & 0 \\ 0 & -S/2 & 0 & 0 \\ 0 & 0 & -I/2 & 0 \\ 0 & C_{cl,z} S & D_{cl,zw} & -\gamma^2 I/2 \end{bmatrix} < 0 \tag{17b}$$

$$S = S^T = \begin{bmatrix} S_{11} & S_{12} \\ S_{12}^T & S_{22} \end{bmatrix} > 0 \tag{17c}$$

$$R_{11} - S_{11} > 0 \tag{17d}$$

$$\begin{bmatrix} 1/s^2 & Z_i \\ Z_i^T & R_{11} \end{bmatrix} \geq 0, \quad i = 1, \ldots, n_u. \tag{17e}$$

Proof: The proof exactly coincides with the proof of Grimm et al. (2003, Proposition 2) characterising the continuous-time counterpart. □

Based on Propositions 1 and 2, the following theorem gives suitable conditions for the feasibility of the conditions in Theorem 4.

Theorem 5: *The following properties hold*:

(1) *The feasibility conditions for a static anti-windup compensator guaranteeing a* (*large enough*) *global finite l_2 gain are satisfied if and only if there exists a matrix R which is a solution to the LMI problem*

$$A_p R_{11} A_p^T - R_{11} < 0$$
$$A_{\mathrm{cl}} R A_{\mathrm{cl}}^T - R < 0$$
$$R = R^T = \begin{bmatrix} R_{11} & R_{12} \\ R_{12}^T & R_{22} \end{bmatrix} > 0. \tag{18}$$

(2) *The feasibility conditions for a plant-order anti-windup compensator guaranteeing a* (*large enough*) *global finite l_2 gain are satisfied if and only if A_p is Schur stable.*
(3) *If conditions* (15) *are feasible with $n_{\mathrm{aw}} = n_1$, then they are feasible with any $n_{\mathrm{aw}} \geq n_1$.*
(4) *If conditions* (15) *are feasible with $n_{\mathrm{aw}} \geq n_p$, then they are feasible with $n_{\mathrm{aw}} = n_p$.*
(5) *The feasibility conditions for a plant-order anti-windup compensator guaranteeing a* (*large enough*) *regional finite l_2 gain are satisfied if and only if $(A_p, B_{p,u})$ is a stabilisable pair.*

Proof: The proof of items 1–4 can be carried out in the same way as the proof of Grimm et al. (2003, Theorem 3).

The remaining item follows from the observation that the parameter Z in condition (17a) can be arbitrary (as long as s can be made arbitrarily small in (17e). Then selecting $Z = K_p R_{11}$ where K_p is a stabilising gain for the plant, and selecting R_{11} as the inverse of a suitable quadratic Lyapunov function for the stabilised plant is sufficient to complete the proof. □

An implication of Theorem 5 is that not only does the anti-windup construction always admit a solution choosing $n_{\mathrm{aw}} = n_p$, but also given a certain optimal performance (either in terms of a guaranteed l_2 gain, a guaranteed stability region or a guaranteed reachable set) achievable by a solution of any order $n_{\mathrm{aw}} \geq n_p$, then by item 4 of the theorem, this same performance is achievable by an anti-windup compensator of order n_p. Hence, the restriction that the anti-windup compensator order is n_p does not restrict the minimum achievable performance level.

Moreover, item 1 of Theorem 5 implies that, in many situations, static anti-windup compensation does not provide a feasible solution to this anti-windup problem. Indeed, condition (18) corresponds to requiring the existence of a 'quasi' common quadratic Lyapunov function between the open-loop plant and the unconstrained closed-loop system. In particular, if the unconstrained controller is static ($R_{11} = R$), it exactly requires a common quadratic Lyapunov function. In the general case of a dynamic unconstrained controller, it is a generalisation of this requirement based on the fact that the size of the unconstrained closed-loop system is larger than the size of the open-loop plant.

3.4 *LMI-based anti-windup synthesis*

Although the results in § 3.2 provide natural conditions for the existence of an anti-windup compensator achieving a certain performance level for the closed-loop system in Figure 2, they do not provide tools for the construction of such a compensator. In this section, based on a solution (R, S, Z, γ^2) arising from Theorem 4 or Proposition 1 or 2, we give a procedure to construct a state-space representation of an anti-windup compensator that guarantees the stability and performance properties listed in Theorem 4. The effectiveness of the procedure is then formally stated in Theorem 6.

To suitably describe the procedure for the construction of the anti-windup compensator, we will first introduce an equivalent representation for the anti-windup closed-loop system (1), (2), (5) and (6) represented in Figure 2. By stacking the plant and the controller states into a single state vector $x_{\mathrm{cl}} := [x_p^T \quad x_c^T]^T \in \mathbb{R}^{n_{\mathrm{cl}}}$, with $n_{\mathrm{cl}} := n_p + n_c$, the anti-windup closed loop system can be written as shown in Figure 4. The dynamics of the subsystem $\mathcal{H}$ in Figure 4 is given by

$$\mathcal{H}\begin{cases} x_{\mathrm{cl}}^+ &= A_{\mathrm{cl}}x_{\mathrm{cl}} + B_{\mathrm{cl},w}w + B_{\mathrm{cl},q}q + B_{\mathrm{cl},v}v \\ z &= C_{\mathrm{cl},z}x_{\mathrm{cl}} + D_{\mathrm{cl},zw}w + D_{\mathrm{cl},zq}q + D_{\mathrm{cl},zv}v \\ y_c &= C_{\mathrm{cl},y}x_{\mathrm{cl}} + D_{\mathrm{cl},yw}w + D_{\mathrm{cl},yq}q + D_{\mathrm{cl},yv}v, \end{cases} \tag{19}$$

where the matrices A_{cl}, $B_{\mathrm{cl},w}$, $B_{\mathrm{cl},q}$, $B_{\mathrm{cl},v}$, $C_{\mathrm{cl},z}$, $D_{\mathrm{cl},zw}$, $D_{\mathrm{cl},zq}$, $D_{\mathrm{cl},zv}$, $C_{\mathrm{cl},y}$, $D_{\mathrm{cl},zw}$, $D_{\mathrm{cl},yq}$ and $D_{\mathrm{cl},yv}$ are of appropriate dimensions and only depend on the matrices of the plant (1) and of the controller (2).

Based on the linear system (19), we can formalise a procedure for the construction of a static [or plant order] anti-windup compensator. The construction of an intermediate order anti-windup compensator involves solving the non-linear feasibility conditions of Theorem 4, which is a very hard numerical problem because it involves a rank constraint. Some results in this direction that apply to the continuous-time case, but may be generalised to the discrete-time case, can be found, e.g. in Galeani, Massimetti, Teel, and Zaccarian (2006), Turner and Postlethwaite (2004) and references therein. Whenever a solution to the non-linear feasibility conditions is available, the controller construction corresponds to a slight generalisation of the plant-order case addressed below, which is omitted here for clarity of exposition.

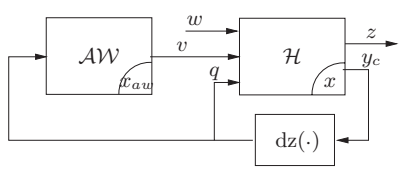

Figure 4. An equivalent representation of the anti-windup closed-loop system.

Procedure 1: Anti-windup compensator construction.

Step 1. *Solve the feasibility LMIs*. Find a solution (R_{11}, S, Z, γ^2) $[(R, Z, \gamma^2)]$ to the feasibility LMIs of Proposition 1 [Proposition 2].

Step 2. *Construct the matrix Q*. Define the matrices $R := \left[\begin{smallmatrix} R_{11} & S_{12} \\ S_{12}^T & S_{22} \end{smallmatrix}\right]$ and $N \in \mathbb{R}^{(n_p+n_c)\times n_{\mathrm{aw}}}$ as a solution of the following equation:

$$RS^{-1}R - R = NN^T. \tag{20}$$

Since R and S are invertible and by the feasibility conditions, $RS^{-1}R - R$ is positive semidefinite and of rank n_{aw}. Hence there always exists a matrix N satisfying Equation (20). Define the matrix $N \in \mathbb{R}^{(n_p+n_c)\times n_{\mathrm{aw}}}$ as $M := I + N^T R^{-1} N$. Finally, define the matrix $Q \in \mathbb{R}^{n\times n}$ $(n = n_p + n_c + n_{\mathrm{aw}})$ as

$$Q := \begin{bmatrix} R & N \\ N^T & M \end{bmatrix}. \tag{21}$$

[Define the matrix $Q = R$.]

Step 3. *Build necessary matrices*. Construct the matrices $A_0 \in \mathbb{R}^{n\times n}$, $B_0 \in \mathbb{R}^{n\times n_u}$, $C_{y0} \in \mathbb{R}^{n_u\times n}$, $D_{yq0} \in \mathbb{R}^{n_u\times n_u}$, $C_{z0} \in \mathbb{R}^{n_z\times n}$, $D_{zq0} \in \mathbb{R}^{n_z\times n_u}$, $B_w \in \mathbb{R}^{n\times n_{\mathrm{w}}}$, $D_{zw} \in \mathbb{R}^{n_z\times n_{\mathrm{w}}}$ and $D_{yw} \in \mathbb{R}^{n_u\times n_{\mathrm{w}}}$ as

$$\left[\begin{array}{c|c|c} A_0 & B_{q0} & B_w \\ \hline C_{y0} & D_{yq0} & D_{yw} \\ \hline C_{z0} & D_{zq0} & D_{zw} \end{array}\right] = \left[\begin{array}{cc|c|c} A_{\mathrm{cl}} & 0 & B_{cl,q} & B_{cl,w} \\ 0 & 0 & 0 & 0 \\ \hline C_{cl,y} & 0 & D_{cl,yq} & D_{cl,yw} \\ \hline C_{cl,z} & 0 & D_{cl,zq} & D_{cl,zw} \end{array}\right].$$

[Define the matrices above by removing the second block row and block column of zeros from the right-hand side of the above equation.]

Step 4. *Anti-windup compensator LMI*. Based on Steps 2 and 3, construct the matrices $H \in \mathbb{R}^{(n_{\mathrm{aw}}+n_c+n_u)\times(2n+n_u+n_w+n_z)}$, $\Psi_R \in \mathbb{R}^{(2n+n_u+n_w+n_z)\times(2n+n_u+n_w+n_z)}$ and $G_U \in \mathbb{R}^{(n_{\mathrm{aw}}+n_u)\times(2n+n_u+n_w+n_z)}$ as follows:

$$\Psi_R = \mathrm{He}\begin{bmatrix} -\frac{Q}{2} & 0 & -Y^T & 0 & 0 \\ A_0Q & -\frac{Q}{2} & B_{q0}U & B_w & 0 \\ C_{y0}Q & 0 & D_{yq0}U - U & D_{yw} & 0 \\ 0 & 0 & 0 & -\frac{I}{2} & 0 \\ C_{z0}Q & 0 & D_{zq0}U & D_{zw} & -\frac{\gamma^2 I}{2} \end{bmatrix},$$

$$H = \left[\begin{array}{c|cc|c|c|c} 0 & 0 & I_{n_{\mathrm{aw}}} & 0 & 0 & 0 \\ 0 & B_{cl,v}^T & 0 & D_{cl,yv}^T & 0 & D_{cl,zv}^T \end{array}\right]$$

$$G_U = \left[\begin{array}{cc|c|c|c|c} N^T & M & 0 & 0 & 0 & 0 \\ 0 & 0 & I_{n_u} & 0 & 0 & 0 \end{array}\right]$$

where $Y \in \mathbb{R}^{n_u \times n_p + n_c + n_{\mathrm{aw}}}$ is defined as $Y := \begin{bmatrix} Z & ZR_{11}^{-1}R_{12} & ZR_{11}^{-1}N_1 \end{bmatrix}$ (where N_1 is the upper block of the matrix N) $[Y := Z \quad ZR_{11}^{-1}R_{12}]$. Finally, solve the LMI

$$\Psi_R + G_U^T \Lambda_U^T H + H^T \Lambda_U G_U < 0 \tag{22}$$

in the unknowns $\Lambda_U \in \mathbb{R}^{(n_{\mathrm{aw}}+n_c+n_u)\times(n_{\mathrm{aw}}+n_u)}$ and $U \in \mathbb{R}^{n_u \times n_u}$, $U > 0$ diagonal, and compute the matrices of the anti-windup compensator (6) as follows:

$$\begin{bmatrix} A_{\mathrm{aw}} & B_{\mathrm{aw}} \\ C_{\mathrm{aw}} & D_{\mathrm{aw}} \end{bmatrix} = \Lambda_U \begin{bmatrix} I & 0 \\ 0 & U^{-1} \end{bmatrix} \tag{23}$$

[compute the static anti-windup gain as $D_{\mathrm{aw}} = \Lambda_U U^{-1}$].

Theorem 6: *Given the plant $\mathcal{P}$, the controller $\mathcal{C}$, an integer n_{aw} and a solution to the conditions in Theorem* 4, *the LMI* (22) *constructed according to Procedure* 1 *is guaranteed to be solvable for Λ. Furthermore, the solution Λ defines the matrices of a linear anti-windup compensator* (6) *of order n_{aw} that guarantees the stability and performance properties characterised in Theorem* 4.

Proof: See § 5.2. □

Remark 6: (A step towards strictly proper anti-windup design) In many practical cases it is of interest to design anti-windup compensators that are strictly proper (namely $D_{\mathrm{aw}} = 0$) or, less restrictively, that do not result in a non-linear algebraic loop (namely, $D_{\mathrm{aw}} = \left[\begin{smallmatrix} D_{\mathrm{aw},1} \\ 0 \end{smallmatrix}\right]$). In these cases, a possible way to tackle this goal is to augment the LMIs (22) with an extra constraint imposing $[0 \;\; I_{n_c+n_u}]\Lambda_U \left[\begin{smallmatrix} 0 \\ I_{n_u} \end{smallmatrix}\right] = 0$ and $[0 \;\; I_{n_u}]\Lambda_U \left[\begin{smallmatrix} 0 \\ I_{n_u} \end{smallmatrix}\right] = 0$, respectively. Of course the arising LMI constraints are not guaranteed to be solvable by the theory, but in many practical cases a solution can be found with this technique which results in an effective practical trick for designing simpler compensators. Generalising this idea, one might be interested in solving (22) while minimising the size of some part of the compensator matrix $\Lambda := \left[\begin{smallmatrix} A_{\mathrm{aw}} & B_{\mathrm{aw}} \\ C_{\mathrm{aw}} & D_{\mathrm{aw}} \end{smallmatrix}\right]$. An example of this which guarantees that the maximum singular value of the compensator matrix Λ is smaller than a constant α could be to impose $U > I$ and augmenting (22) with

$$\begin{bmatrix} \alpha I & \Lambda_U \\ \Lambda_U^T & \alpha I \end{bmatrix} \geq 0,$$

which guarantees by Schur complement that $\Lambda^T \Lambda < \Lambda_U^T \Lambda_U \leq \alpha^2 I$. Clearly, one can solve the arising constraints while minimising α thereby obtaining a selection of Λ with very small maximum singular value.

4. Simulation example

Consider the digital control system illustrated in Franklin, Powell, and Emami-Naeini (2002), resembling the key features of the angular position regulation for a satellite where the three axes are completely decoupled, so that one can concentrate on only one axis and implement three decoupled controllers. The simplified plant is described by the following transfer function:

$$P(z) = \frac{T^2}{2} \frac{z+1}{(z-1)^2},$$

where we choose a sampling time of $T = 0.1$ s. The system is unstable but not exponentially unstable, because the two poles are coincident in $z = 1$ (this obviously corresponds to a discretised double integrator). Note that global exponential stability cannot be obtained for this plant, but only regional exponential stability is achievable, regardless of the unconstrained controller selection.

According to Franklin et al. (2002), we pick the unconstrained controller as

$$C(z) = \frac{0.692}{T^2/2} \frac{z - 0.63}{z + 0.44}.$$

A possible state-space representation of the system obtained by using Matlab's `ss` command is given by

$$\left[\begin{array}{c|c|c} A_p & B_{p,u} & B_{p,w} \\ \hline C_{p,y} & D_{p,yu} & D_{p,yw} \\ \hline C_{p,z} & D_{p,zu} & D_{p,zw} \end{array}\right] = \left[\begin{array}{cc|c|c} 2 & -0.5 & 0.125 & 0 \\ 2 & 0 & 0 & 0 \\ \hline 0.04 & 0.02 & 0 & 0 \\ \hline -0.04 & -0.02 & 0 & 1 \end{array}\right]$$

$$\left[\begin{array}{c|c|c} A_c & B_{c,y} & B_{c,w} \\ \hline C_c & D_{cy} & D_{cw} \end{array}\right] = \left[\begin{array}{c|c|c} -0.44 & -16 & 16 \\ \hline -9.2555 & -138.4 & 138.4 \end{array}\right],$$

where the performance output z has been selected as the tracking error $r - y$ with $w = r$ being the reference input.

For our simulations, we select the reference input w as a step occurring at time $t = 0$ and rising from 0 to 50. To induce desirable unconstrained behaviour, the reference is fed into a feedforward low-pass discrete-time filter whose transfer function is

$F(z) = \frac{0.01489}{z-0.9851}$. The use of this prefilter resembles the typical actions taken in satellite control where the reference input is selected based on the computation of a desired position, velocity and acceleration profile. Therefore, simply using step references would be unreasonable and would induce unsatisfactory responses even without saturation.

The control input u is assumed to be bounded between ± 1. Since the plant is not exponentially stable, then global results cannot be obtained by our approach because global exponential stability is not achievable from the bounded input u. Therefore, we seek static and plant-order anti-windup guaranteeing a suitable regional performance optimised for some fixed value s of the l_2 norm of w.

A static anti-windup compensator has been first determined following Procedure 1 with $s = 0.01234$. The arising static anti-windup compensator, which induces an optimised regional l_2 gain $\gamma = 355.29$, is given by $D_{\mathrm{aw}} = \left[\begin{smallmatrix} -0.83419 \\ -16.578 \end{smallmatrix}\right]$. Then a dynamic anti-windup compensator has been determined (once again using Procedure 1) with $s = 0.14$. The arising compensator, inducing an optimised regional l_2 gain $\gamma = 156.34$ is given by

$$\left[\begin{array}{c|c} A_{\mathrm{aw}} & B_{\mathrm{aw}} \\ \hline C_{\mathrm{aw}} & D_{\mathrm{aw}} \end{array}\right] = \left[\begin{array}{cc|c} 0.22162 & 0.35832 & -0.0089518 \\ 0.45986 & 0.74282 & -0.018416 \\ \hline 4.1095 & 6.9548 & -0.20242 \\ 33.703 & 61.119 & -0.26949 \end{array}\right].$$

Figure 5 shows the closed-loop responses with and without anti-windup. From the dotted curves (saturation but no anti-windup), it is evident that the closed-loop system is prone to windup. As compared to the unconstrained closed-loop response (dashed), unacceptable oscillations are experienced at the plant output (see the upper plot). Unfortunately, for this example, static anti-windup does not seem to do the job in a satisfactory way. Indeed the arising response (dash-dotted) is oscillatory as well and not much different from the saturated one, just commented. Conversely, the plant order compensation leads to the solid response which is desirable because, in spite of a slight initial performance loss during saturation, the performance recovery is achieved by an almost bang–bang behaviour. The unconstrained response is recovered by the anti-windup closed loop already during the transient phase, which denotes a successful anti-windup action. This type of response is quite consistent with references of different magnitude although extremely large values of the reference eventually lead to some undesirable overshoots also on the system with plant-order anti-windup. To overcome those overshoots it would be necessary to design anti-windup compensators with larger values of s, so that the larger signals behaviour is optimised.

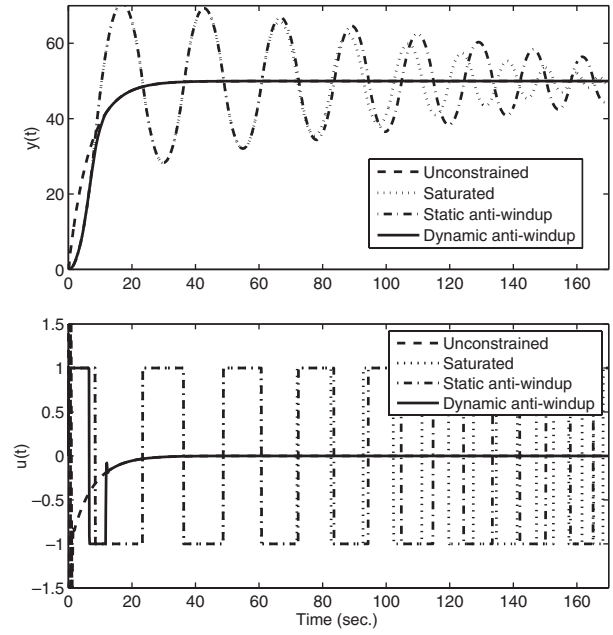

Figure 5. Plant output y (upper plot) and plant input u (lower plot) without saturation (dashed), without anti-windup (dotted), with static anti-windup (dash-dotted) and with plant-order anti-windup (solid).

The reason why static anti-windup (dash-dotted) is unsatisfactory in the simulations of Figure 5 can be found in the fact that the lesser degrees of freedom available to this simpler compensation scheme are insufficient to recover the windup effects for this example. This limitation can be guessed already in the design phase when feasibility is only obtained upon severely reducing the value of s, which suggests that the region where the arising anti-windup is effective becomes too small. However, an even more convincing argument can be carried out by looking at the non-linear gain estimates from r to z corresponding to the three non-linear closed loops (saturated without anti-windup, with static anti-windup and with plant-order anti-windup). These are compared in Figure 6, where the dotted curve corresponds to the saturated closed loop without anti-windup. From the curve it appears that for values of s larger than (roughly) 10^{-2}, there's no finite l_2 gain estimate via our quadratic Lyapunov functions. Static anti-windup (dashed curve) does not seem to improve the situation much, as a matter of fact, the regional l_2 gain appears improved below the above mentioned threshold of 10^{-2}, but no significant extension of this threshold appears to be possible. Conversely, dynamic anti-windup shows a significant extension of the region where the l_2 gain estimate can be computed (solid curve). This anticipates the desirable simulation response experienced in Figure 5.

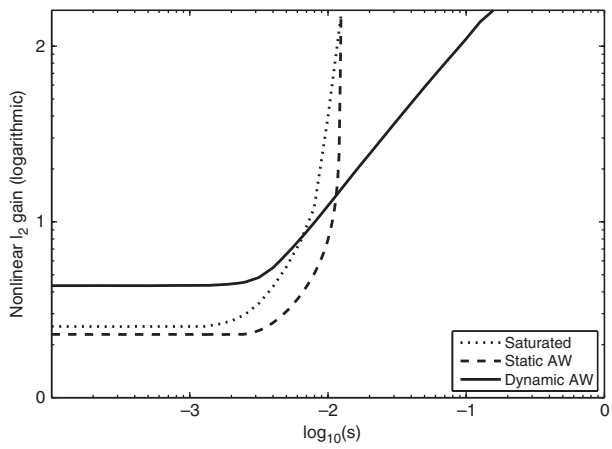

Figure 6. Non-linear l_2 gains estimates of the saturated closed loop without anti-windup (dotted), of the closed loop with static anti-windup (dashed) and of the closed loop with plant-order anti-windup (solid).

Note that the dynamic anti-windup closed loop induces a slight deterioration of the small signal l_2 gain (the scale is logarithmic). This seems to be the price to be paid for achieving the desirable large signal behaviour.

Finally, it must be acknowledged that the estimates in Figure 6 are pretty conservative because they rely on the use of simple quadratic Lyapunov functions. More accurate estimates could be obtained by relying on non-convex nonquadratic Lyapunov conditions (this could be done, for example, by extending to the discrete-time case the results in Hu et al. (2006)). However, we still believe that a comparative look at the gain estimates in Figure 6 gives a rough idea of where certain solutions work better than other ones. For instance, if we run the simulation of Figure 5 after making the reference 10 times smaller, the four curves (unconstrained, saturated, static AW and dynamic AW) are all very close to each other but the response using dynamic anti-windup is the one that deviates the most from the unconstrained response. This is somehow suggested by the fact that in Figure 6, the closed loop leading to the worst small signals gain is the one characterising the dynamic anti-windup closed loop.

5. Proof of the main result

5.1 *Proof of the analysis theorems: Theorems 1, 2 and 3*

The proofs are based on a Lyapunov technique showing that under the generalised sector condition used in e.g. Gomes da Silva and Tarbouriech (2001), Hu, Lin, and Chen (2002), Hu, Lin, and Chen (2002) condition (10) implies a decrease in the increment of the Lyapunov function along the dynamics of systems (8) and (9). Moreover, the condition encompasses the extra performance properties stated in the three theorems. The following lemma (see Gomes da Silva and Tarbouriech (2001), Hu et al. (2002), Hu et al. (2002)) characterises the generalised sector condition used here.

Lemma 2: *Let $H \in \mathbb{R}^{n_u \times n}$ be given. For every diagonal matrix $M \in \mathbb{R}^{n_u \times n_u}$, $M > 0$, we have for all $y_c \in \mathbb{R}^{n_u}$*

$$-2q^T M(q - y_c + Hx) \geq 0, \ \forall x \in \mathcal{L}(H). \tag{24}$$

Proof of Theorem 2: Well-posedness can be shown following the same argument as in the continuous-time counterpart (see e.g. Grimm et al. (2003), Theorem 1) based on the inequality $-2U + D_{yq}U + UD_{yq}^T < 0$, which is implied by (10). For the rest of the proof, let $P = Q^{-1}$, $H = YQ^{-1}$ and $M = U^{-1}$. Left- and right-multiplying of Equation (12) by $\mathrm{diag}\{P, P, M, I\}$ and taking a Schur complement, we have

$$\bar{\Phi} = \mathrm{He}\begin{bmatrix} \dfrac{A^T PA - P}{2} & A^T PB_q - H^T M^T & 0 \\ MC_y & MD_{yq} - M + \dfrac{B_q^T PB_q}{2} & MD_{yw} \\ B_w^T PA & B_w^T PB_q & \dfrac{B_w^T PB_w - I}{2} \end{bmatrix} \leq 0. \tag{25}$$

Also, (11) is equivalent to

$$\begin{bmatrix} \dfrac{1}{s^2} & H_i \\ H_i^T & P \end{bmatrix} \quad \forall i = 1, 2, \ldots n_u,$$

which implies that $\mathcal{E}(\mathcal{P}/s^2) \subset \mathcal{L}(H)$ (see e.g. Boyd, El Ghaoui, Feron, and Balakrishnan (1994)).

Consider the quadratic function $V(x) = x^{\mathrm{T}} Px$. We would like to exploit Lemma 2 to show that

$$\Delta V - w^T w = (x^+)^T Px^+ - x^T Px - w^T w \leq 0 \tag{26}$$

for all $x \in \mathcal{E}(P/s^2) \subset \mathcal{L}(H)$, where $x^+ = Ax + B_q q + B_w w$. To this aim, defining $\xi = [x^T \ \ q^T \ \ w^T]^T$ and adding (24) to (26) we get after some computations

$$\Delta V - w^T w \leq \xi^T \bar{\Phi} \xi \leq 0.$$

Hence we have that $\Delta V \leq w^T w$ for all $x \in \mathcal{E}(P/s^2)$. Now suppose $x(0) = 0$ and $\|w\|_2^2 \leq s^2$. Then for any $t_0 > 0$, as long as $x(t) \in \mathcal{E}(P/s^2)$ for all $t \in [0, t_0]$, we have

$$V(x(t_0)) \leq \sum_{\tau=0}^{t_0} w^T(\tau) w(\tau) \leq s^2,$$

namely, $x(t_0) \in \mathcal{E}(P/s^2)$. On the other hand, if there exists $t_0 > 0$ such that $V(x(t)) \le s^2$ for all $t \in [0, t_0 - 1]$ and $V(x(t_0)) = s^2$ then we must have

$$\sum_{\tau=t_0}^{\infty} w^T(\tau) w(\tau) = 0$$

and $\Delta V(x(t)) \le 0$ for all $t > t_0$. Hence $V(x(t)) \le s^2$ for all $t > t_0$. Therefore, we conclude that $x(t) \in \mathcal{E}(P/s^2)$ for all $t > 0$.

Proof of Theorem 1: The proof follows a similar approach to the previous one. Taking a Schur complement in the last rows and columns of (10), we obtain

$$\bar{\Phi} + \begin{bmatrix} C_z^T \\ D_{zq}^T \\ D_{zw}^T \end{bmatrix} \frac{\begin{bmatrix} C_z & D_{zq} & D_{zw} \end{bmatrix}}{\gamma^2} < 0. \tag{27}$$

Repeating the calculations of the proof of Theorem 2 using the quadratic function $V(x) = x^T P x$, we obtain from the sector condition (24) and from (27) that

$$\Delta V + \frac{z^T z}{\gamma^2} < w^T w, \ \forall\, x \in \mathcal{E}(P/s^2) \setminus \{0\}.$$

Summing and using $\mathcal{E}(P/s^2) \subset \mathcal{L}(H)$, it follows that

$$\|z\|_2 \le \gamma \|w\|_2, \ \forall w(\cdot) \quad \text{s.t. } \|w\|_2 \le s.$$

Proof of Theorem 3: Following the same approach as in the proof of Theorem 2, by the sector condition (24) and from Equation (14) it follows that for the quadratic function $V(x) = x^T P x$,

$$\Delta V(x) < 0, \ \forall x \in \mathcal{E}(P) \setminus \{0\}.$$

5.2 *Proof of Theorems 4 and 6*

The proof is adapted from the proof of Grimm et al. (2003, Theorem 5) also using the generalised sector conditions exploited in Hu et al. (2008) to obtain regional constructions in the continuous-time case. To avoid overlap, we will outline the main idea and point out the differences between the situation in this article and that in Grimm et al. (2003). The following simple fact, which can be shown by applying a Schur complement twice, will be needed to complete the proof:

Fact 1: Consider matrices A_{11}, A_{12} and $B = \left[\begin{smallmatrix} B_{11} & B_{12} \\ B_{12}^T & B_{22} \end{smallmatrix}\right] > 0$ of suitable dimensions. Then the following holds:

$$\left[\begin{array}{c|c} A_{11} & \begin{bmatrix} A_{12} & 0 \end{bmatrix} \\ \hline \begin{bmatrix} A_{12}^T \\ 0 \end{bmatrix} & B^{-1} \end{array}\right] > 0 \ \Leftrightarrow \begin{bmatrix} A_{11} & A_{12} \\ A_{12}^T & B_{11}^{-1} \end{bmatrix} > 0.$$

The proof of Theorems 4 and 6 is carried out establishing that conditions (15) imply that conditions (10) and (11) of Theorem 1 are satisfied by the anti-windup compensator arising from (22) in Procedure 1. In particular, we will first show that (15a), (15b), (15e) and (15f) are equivalent to the feasibility of (10) and then show that (15i) implies the feasibility of (11). Then all the items of the theorem follow in a straightforward way from Theorems 1–3.

Feasibility of (10). Let Ψ_R, H and G_U be formed as in Step 4 of Procedure 1. Similar to the computation in Grimm et al. (2003), it can be verified that the left-hand side of (22) satisfies the following equation:

$$\Phi = \Psi_R + G_U^T \Lambda_U^T H + H^T \Lambda_U G_U, \tag{28}$$

where Φ is given in (10) (note that, as compared to the definitions of G and Λ in Grimm et al. (2003), G_U and Λ_U are different here, even though their product satisfies $\Lambda G = \Lambda_U G_U$. Indeed, in Procedure 1, the term U is factored out as an extra unknown).

Using Grimm et al. (2003, Lemma 5) (projection lemma), the existence of Λ_U satisfying $\Phi < 0$ is equivalent to the feasibility of

$$W_H^T(\Psi_R) W_H < 0 \tag{29}$$

$$W_{G_U}^T(\Psi_R) W_{G_U} < 0, \tag{30}$$

where W_{G_U}, W_H can be any matrices whose columns form bases of the null space of G_U and H.

A possible selection of W_H (paralleling the continuous-time one constructed in Grimm et al. (2003)) is

$$W_H = \begin{bmatrix} I_n & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & I_{n_p} & 0 & 0 & -B_{p,u} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & I_{n_w} & 0 \\ 0 & 0 & 0 & 0 & -D_{p,zu} & 0 & I_{n_z} \end{bmatrix}^T,$$

where the first 4×4 blocks add up to a total dimension of $(n + n_p + n_w + n_z) \times 2n$. Assume that Q is partitioned as

$$Q = \begin{bmatrix} R & N \\ N^T & M \end{bmatrix}, \ R = \begin{bmatrix} R_{11} & R_{12} \\ R_{12}^T & R_{22} \end{bmatrix}, \tag{31}$$

where $R \in \mathbb{R}^{(n_p+n_c)\times(n_p+n_c)}$, $R_{11} \in \mathbb{R}^{n_p \times n_p}$. Similar to Grimm et al. (2003), it can be verified after tedious calculations that

$$W_H^T(\Psi_R) W_H = \left[\begin{array}{ccc|ccc} -R_{11} & B_{p,w} & 0 & A_p + B_{pu} Z & 0 & 0 \\ B_{p,w}^T & -I_{n_w} & D_{p,zw}^T & 0 & 0 & 0 \\ 0 & D_{p,zw} & -\gamma^2 I_{n_z} & C_{p,z} + D_{p,zu} \dot{Z} & 0 & 0 \\ \hline & \star & & & -Q^{-1} & \end{array}\right]$$

where

$$Z = Y\begin{bmatrix} I_{n_p} \\ 0 \\ 0 \end{bmatrix} \in \mathbb{R}^{n_u \times n_p},$$

and $\star$ denotes symmetric entries. By Fact 1, it follows that (29) is equivalent to (15a).

A possible selection of W_{G_U} (paralleling the continuous-time one constructed in Grimm et al. (2003)) is $W_{G_U} = \bar{T}^{-1} W_{G\circ}$ with $\bar{T} = \{Q, Q, I, I, I\}$ and

$$W_{G\circ} = \begin{bmatrix} I_{n_{\rm cl}} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I_{n_{\rm cl}} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & I_{n_{\rm aw}} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & I_{n_w} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & I_{n_z} \end{bmatrix}.$$

If we let $\bar{\Psi}_R = \bar{T}^{-1}\Psi_R\bar{T}^{-1}$, then $W_G^T\Psi_R W_G = W_{G\circ}^T\bar{\Psi}_R W_{G\circ}$. Partition P as $P = \left[\begin{smallmatrix} S^{-1} & P_{12} \\ P_{12}^T & P_{22} \end{smallmatrix}\right]$ with $S \in \mathbb{R}^{(n_p+n_c)\times(n_p+n_c)}$, then, after tedious calculations, it can be verified that $W_{G_U}^T\Psi_R W_{G_U} < 0$ is equivalent to

$$\left[\begin{array}{cc|ccc} \multicolumn{2}{c|}{-P^{-1}} & B_{cl,w} & 0 & A_{\rm cl} \\ & & 0 & 0 & 0 \\ \hline B_{cl,w}^T & 0 & -I_{n_w} & D_{cl,zw}^T & 0 \\ 0 & 0 & D_{cl,zw} & -\gamma^2 I_{n_z} & C_{cl,z} \\ A_{\rm cl}^T & 0 & 0 & C_{cl,z}^T & -S^{-1} \end{array}\right] < 0.$$

Therefore, applying Fact 1 and right- and left-multiplying by diag $\{I, I, I, S\}$, it follows that (30) is equivalent to (15b).

Note that (15a) and (15b) are stated in terms of R_{11} and S, which are constrained by

$$\begin{bmatrix} R & N \\ N^T & M \end{bmatrix} \times \begin{bmatrix} S^{-1} & P_{12} \\ P_{12}^T & P_{22} \end{bmatrix} = QP = I. \qquad (32)$$

By Grimm et al. (2003, Lemma 6) (see also Packard (1994)), (32) is satisfied if constraint (15e) and (15f) is satisfied.

Feasibility of (11). Assume that (15i) is satisfied and define $K_{pi} := Z_i R_{11}^{-1}$. Then, by Schur complement, and the definition of Q in (31),

$$\begin{aligned} 0 \le \frac{1}{s^2} - Z_i R_{11}^{-1} Z_i^T &= \frac{1}{s^2} - K_{pi} R_{11} K_{pi}^T \\ &= \frac{1}{s^2} - \begin{bmatrix} K_{pi} & 0 & 0 \end{bmatrix} Q \begin{bmatrix} K_{pi} & 0 & 0 \end{bmatrix}^T \\ &= \frac{1}{s^2} - Y_i Q^{-1} Y_i^T, \end{aligned}$$

where, according to the selection in Step 4 of Procedure 1,

$$\begin{aligned} Y_i &= \begin{bmatrix} K_{pi}R_{11} & K_{pi}R_{12} & K_{pi}N_1 \end{bmatrix} \\ &= \begin{bmatrix} Z_i & Z_i R_{11}^{-1} R_{12} & Z_i R_{11}^{-1} N_1 \end{bmatrix}. \end{aligned}$$

Finally, by Schur complement, the last inequality above easily transforms into (11).

Theorem 6 trivially follows from the fact that the right-hand side of (28) coincides with the left-hand side of (22).

6. Conclusions

In this article we have characterised full-authority linear anti-windup augmentation for linear discrete-time control systems in the presence of input saturation. The approach parallels the corresponding continuous-time results and allows for regional and global quadratic stability and performance guarantees. LMI conditions have been given for optimal anti-windup construction for the special cases of static and plant-order anti-windup augmentation. The effectiveness of the proposed approach has been illustrated on a simulation example.

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Notes

1. For simplicity, we restrict the attention in this article to unit saturation functions, defined as the scalar saturation with unity limits, namely $\sigma(w) := \max\{-1, \min\{1, w\}\}$. However, the approach can be easily generalised to the case of general symmetric saturations.
2. The LMI eigenvalue problem (see e.g. Boyd et al. (1994, page 10)) is to minimise a linear function subject to an LMI constraint (or to determine that the constraint is unfeasible).
3. Eigenvalue problems can be efficiently solved in polynomial time by using commercial software such as the LMI Control Toolbox (Gahinet, Nemirovski, Laub, and Chilali 1995) of MATLAB.

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